

Vendors Warning Walkthrough — How the HIGH Alert Is Generated

Essential calculations only (Vendors / 18th transaction type)

July 6, 2026

Vendors Warning Walkthrough

This note traces **only** the calculations that turn today's Vendors signal into the dashboard warning **8.9%** · $z \approx -7.28$ · **HIGH**, ranked #1. Values are taken from `NGAME_Fraud_Analysis_readable_truly_clean.json` and the Vendors row of `NGAME_Training_Matrix.xlsx`.

Given (inputs to the warning path)

Quantity	Symbol	Value
Today's Vendors CPI	$\text{CPI}_{18}^{\text{today}}$	0.90625
Trained mean	μ_{18}	0.994794174882256
Trained std dev	σ_{18}	0.01216178288304404

Step 1 — Percent deviation (dashboard readout: 8.9%)

$$\text{absolute_difference} = |\text{CPI}_{18}^{\text{today}} - \mu_{18}| = |0.90625 - 0.994794174882256| = 0.088544174882256$$

$$\text{percent_deviation} = \frac{\text{absolute_difference}}{\mu_{18}} \times 100 = \frac{0.088544174882256}{0.994794174882256} \times 100 = 8.90\%$$

Step 2 — z -score (dashboard readout: $z \approx -7.28$)

$$z_{18} = \frac{\text{CPI}_{18}^{\text{today}} - \mu_{18}}{\sigma_{18}} = \frac{0.90625 - 0.994794174882256}{0.01216178288304404} = \frac{-0.088544174882256}{0.01216178288304404} = -7.280525868 \Rightarrow z \approx -7.28$$

Negative z : today's CPI is **below** the trained mean (more churn than normal).

Step 3 — Deviation level (HIGH)

Rule	Level	This case
$ z > 3.0$	HIGH	$7.28 > 3.0 \Rightarrow \mathbf{HIGH}$
$2.0 < z \leq 3.0$	MEDIUM	—
$ z \leq 2.0$	LOW	—

Step 4 — Dollar-weighted path (wCPI)

For Vendors, each record has no dollar amount; weights default to \$1, so the dollar-weighted index equals the count-based CPI:

$$\text{wCPI}_{18}^{\text{today}} = 0.90625 = \text{CPI}_{18}^{\text{today}}$$

$$\text{dollar_}z_{18} = -7.280525868 \quad (\text{same as } z_{18})$$

$$\text{divergence} = \left| \text{wCPI}_{18}^{\text{today}} - \text{CPI}_{18}^{\text{today}} \right| = 0$$

`dollar_alarm_level` = "HIGH" (driven by dollar z , not divergence).

Step 5 — Composite alarm and rank

`composite_alarm` = (`deviation_level` $\in$ {HIGH, MEDIUM}) $\vee$ `dollar_alarm` = HIGH $\vee$ `true` = **TRUE**

`rank` = 1 (largest $|z|$ among all 18 transaction types)

Summary chain

Step	Formula	Plugged in	Result
1	$\%dev = CPI - \mu / \mu \times 100$	$0.088544 / 0.994794 \times 100$	8.90%
2	$z = (CPI - \mu) / \sigma$	$-0.088544 / 0.012162$	-7.281
3	$ z > 3 \Rightarrow \text{HIGH}$	$7.28 > 3$	HIGH
4	$\text{wCPI} = \text{CPI}$ (Vendors)	0.90625	dollar HIGH
5	<code>composite_alarm</code>	HIGH $\vee$ <code>dollar_alarm</code>	TRUE, rank 1

Dashboard line: *Vendors (the 18th transaction type) — 8.9% · $z \approx -7.28$ · HIGH, rank #1.*

Formulas per `ngame_churn_comparison_agent.py` (z -score, deviation bands, dollar/composite alarms).